# AI Models for Handwritten Text Recognition: Evaluation Report

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## Background

At the beginning when I was researching on how to satisfy the client's requirements, I identified two major ways to perform Handwritten Text Recognition (HTR): (1) use locally-trained models via manual annotations of the line segmentations and transcription for each spine label using eScriptorium (that runs on the underlying Kraken engine); or (2) incorporate pre-existing models from cloud-based AI providers into my group's product.

I initially went with approach (1), because it was completely free of charge and the whole process will be completely local (i.e. no third-party involvement). However, I soon realised that the process of manually annotating the line segmentation of each word and its transcription in each picture is painstakingly slow. Even if I trained the models across many images, the results weren't promising either.

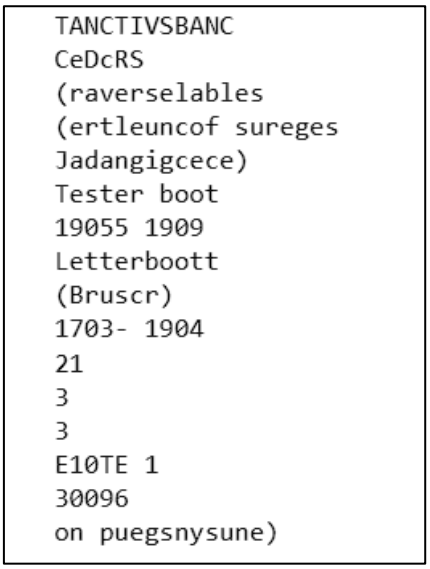



*Figure 1 - Transcription of test_27.jpg using the locally-trained models in approach (1). The models has been trained over 30 images, but there still exists serious confusions between the characters, expecially in the canonical number and keys.*

As such, after conveying my finding to the client and knowing that he has previously attempted to use AI models to perform HTR, I proceeded to evaluate the feasibility of approach (2). For this evaluation, I have chosen Google's *Gemini 3 Flash Preview*, DeepSeek's *V4.1 Flash*, and Anthropic's *Claude Sonnet 5* as my test subjects.

## Methodology

To ensure fairness, each AI model will be fed with the same 26 images (of distinct, individual spine labels and be given the same prompt:

> *"Extract all text from this image exactly as it appears. Preserve line breaks and formatting. Do not summarize or add commentary. Ignore strikethroughs and random numbers in the corner."*

The results were written into a CSV file, which I then converted it to Excel for better formatting. Each row of the Excel file contains the file name of the image, description, input token cost, output token cost, and total token cost (the sum of input and output).

Since these APIs require credits, from which they will deduct the cost (based on the number of tokens used) from the credit balance, I evaluated each AI model based on its input and output token costs and the corresponding price. On top of that, I will make some observations each model's ability to accurately extract the texts, but not so much, since they all performed similarly with a few variations. The client is more than encouraged to look at the text transcriptions for each model in the Excel files to observe the performance, so **I will leave it up to the client to give the verdict on which model is the best while considering the costs.**

It can be said that the main objective of this evaluation is to find the key balance between performance and efficiency; identify a model that performs well, but at the same time does not require too many tokens.

# Models

## Gemini 3 Flash Preview

### Pricing

While Gemini API provides a Free tier for developers with small projects, it unfortunately comes with a few caveats. Firstly, Free users are only allowed to make at most 5 requests per minute (RPM) and up to 20 requests per day (RPD), or in this context, they can only extract texts from 5 images per minute and 20 images for the whole day, or a total of only 140 individual spine labels per week. Furthermore, Gemini explicitly states that they will use Free users' content (i.e. prompts and responses) to improve their products. Therefore, if the client prioritises high efficiency and guarantee to strict privacy, then **Gemini API's Free tier is unfeasible**.

For Gemini API's Paid tier, the standard pricing for the Gemini 3 Flash Preview model is as follows:

- Input price: **0.50 USD per one million tokens**
- Output price: **3.00 USD per one million tokens**

Looking at the results from *gemini_3_flash_token_usage.xlsx*, the average input token cost is **1122.96,** and the average output token cost is **77.69**. Using the above pricing, processing 1,000 images will cost about **0.74 USD in total.**

### Performance

The extracted texts seem to be coherent for all images with minimal noise, with the exception of *test_08.jpg*, which may or may not be caused by a bug in the code.

## DeepSeek V4.1 Flash

### Pricing

DeepSeek's pricing has two sets of rates: **peak rates** (01:00 – 04:00 UTC and 06:00 – 10:00 UTC, Monday through Friday) and **off-peak rates**, which are half of the peak rates. The pricing per one million tokens is as follows:

- Peak rates
    - Input price: **0.30 USD**

    - Output price: **1.20 USD**
- Off-peak rates
    - Input price: **0.15 USD**
    - Output price: **0.60 USD**

Looking at the results from *deepseek_flash_token_usage.xlsx*, the average input token cost is **484.77,** and the average output token cost is **236.23**. Using the above pricing, processing 1,000 images will cost about **0.43 USD during peak hours** and **0.21 USD during off-peak hours**.

### Performance

The text extraction seems to have functioned properly with near-perfection transcription besides a few odd characters, as shown in *test_05.jpg* and *test_26.jpg*.

## Claude Sonnet 5

### Pricing

The pricing for Sonnet 5 API is as follows:

- Input price: **2.00 USD per one million tokens**
- Output price: **10.00 USD per one million tokens**

Looking at the results from *sonnet_5_flash_token_usage.xlsx*, the average input token cost is **908.31,** and the average output token cost is **108.35**. Using the above pricing, processing 1,000 images will cost about **2.90 USD in total.**

# Comparison

(Prices are in USD per one million tokens)

| Cost/Model | DeepSeek V4.1 | Gemini 3 Flash | Claude Sonnet 5 |
|---|---|---|---|
| **Input cost** | 0.30 (peak), 0.15 (off-peak) | 0.50 | 2.00 |
| **Output cost** | 1.20 (peak), 0.60 (off-peak) | 3.00 | 10.00 |
| **Total cost (for 1,000 images)** | 0.43 (peak), 0.21 (off-peak) | 0.74 | 2.90 |